

Coding & Solution

Date	25 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality, implementation, and completeness of the **Emotion Detection & Learning Support Engine**. The project uses Machine Learning, NLP, and Generative AI to detect student emotions from text and provide personalized learning support.

Instructions

1. Ensure the code follows modular programming principles.
2. Use meaningful variable and function names.
3. Follow Python coding standards (PEP-8).
4. Include proper comments where required.
5. Implement exception handling for critical operations.
6. Ensure reusable components.
7. Include setup instructions.
8. Maintain clean project architecture.

Field	Details
Repository Link / URL	GitHub Repository Link
Programming Language(s)	Python 3.10+
Framework(s) Used	Streamlit, Scikit-learn, Google Gemini API, Pandas, NumPy, Plotly
Key Features Implemented	• Emotion Detection using NLP and Machine Learning • Student Emotion Classification • Personalized Learning Recommendations • Google Gemini AI Integration • Interactive Streamlit Dashboard • Emotion Analytics • Session History Tracking • AI-powered Learning Support
Pending / Incomplete Features	• User Authentication • Cloud Database • Voice Emotion Recognition • Multi-language Support • Mobile Application
Setup / Run Instructions	Step 1: Clone Repository Step 2: pip install -r requirements.txt Step 3: Configure Gemini API Key Step 4: streamlit run app.py Step 5: Open the generated local URL

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into reusable functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Proper error handling is implemented	Yes
5	Application executes without critical errors	Yes
6	Source code is maintained using GitHub version control	Yes

Additional Notes / Comments

The application follows a modular architecture with dedicated modules for text preprocessing, emotion detection, machine learning inference, Gemini AI integration, Streamlit UI, analytics, and reporting. The architecture supports future enhancements including multilingual emotion detection, speech-based emotion recognition, cloud deployment, and adaptive learning analytics.